

Any League? Probabilistic Player Archetypes in Women's Basketball

Replicating *Scouting Anyone* in the Liga Femenina Endesa

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Abstract

The probabilistic archetypes of "Scouting Anyone" (MIT Sloan 2026), built on men's leagues, transfer to women's basketball: role players match closely, interior and high-usage roles do not. Applying the paper's exact recipe (Archetypal Analysis on 15 box-score features, eight archetypes) to 142 player-team stints of the Spanish Liga Femenina Endesa 2025/26, scraped and validated from the federation's website, yields eight interpretable archetypes that explain 69.5% of the variation, against 49.6% for K-Means. Three findings stand out. First, role players come in types and stars do not: archetype stability across 20 refits falls with usage (correlation −0.81), from 89% for the Spot-Up Shooter to 26% for the Scoring Wing. Second, the paper's Traditional Playmaker and 3PT Specialist have robust counterparts in the Floor General and the Spot-Up Shooter, while the interior is organised differently and the Combo Creator has no men's equivalent. Third, players are mixtures: the typical main-archetype weight is below 50%. Projected with the frozen model, 30% of players change role between seasons beyond a noise baseline that 10% would exceed by chance.
Keywords: archetypal analysis; women's basketball; player roles; Liga Femenina Endesa; replication

1. Introduction

Positions no longer describe what basketball players do, and this study tests whether a method built to replace them works in women's basketball. A point guard can be a pass-first organiser or a volume scorer; a small forward can play as a rim protector. In the sample studied here, only 49% of small forwards' playing style falls within the wing family of roles.

"Scouting Anyone: Probabilistic Player Archetypes for Any League" (Myrte, Buzzalino and Aymerich, MIT Sloan Sports Analytics Conference 2026) proposes an alternative. It uses Archetypal Analysis (AA) to describe each player as a mixture of extreme playing styles, built only from widely available box-score statistics. The paper fits eight archetypes on seven men's European domestic leagues and on the NBA, and presents the method as valid for any league.

No women's league was tested. This study applies the paper's recipe, with the same 15 features, minutes filter, scaling and number of archetypes, to the Spanish Liga Femenina Endesa, and asks three questions:

1. Do the paper's archetypes exist in a women's league, and where do they differ?
2. Are the archetypes stable enough to trust when the model is refitted?
3. Do they hold when the frozen model is projected onto a new season (2026/27)?

The work also applies the tools of Classes 8 to 10 of the Sports Analytics master's: data collection and wrangling with pandas, data visualisation, PCA, K-nearest neighbours and clustering.

2. Data: building the dataset (notebook 01_dados)

The final dataset covers 296 games and 5,783 player-game rows over two seasons, every one parsed without errors and checked against the federation's official totals. Getting there took several attempts, kept visible in the notebook.

Season	Role in the study	Games	Players	Player-game rows
2025/26, regular season	Training	240	248	4,659
2026/27, rounds 1 to 7	Projection	56	192	1,124

2.1 Choosing the source

The first candidate, Be-Basketball, refused automated requests (HTTP 403). The official website of the Spanish Basketball Federation (FEB) became the source: it is the reference for the league, and its pages are plain HTML tables. Pages are downloaded with `requests` (with a browser User-Agent and a one-second pause between requests), and tables are read with `pd.read_html`, using `extract_links="body"` to keep each player's profile link, which serves as her identifier.

2.2 First approach: team rosters, and why it failed

The first scraper went team by team: read each roster, then each player's season page. Two problems appeared. Eight teams failed with an unpacking error, because a player who had also played the Supercopa had an extra averages table, so the table at a fixed position was the wrong one; the fix was to pick the table by its content (20 columns and a "made/attempted" value in the fifth) and to wrap each player in a try/except. The second problem was fatal: rosters list only current players, so anyone who had left mid-season disappeared. Valencia's player totals fell 103 points short of the team's official total.

2.3 Second approach: box scores from the calendar

The scraper was rebuilt around games instead of rosters. It reads the season calendar, collects every game's link, and parses each box score (`get_games`, `get_game_boxscore`, `get_season_boxscores`). Every player who stepped on court is now counted, whether she is still on the roster or not. Each box-score row is cleaned into 20 named columns; cells such as "5/8" are split into made and attempted, and minutes in "mm:ss" become decimals.

2.4 The past season: a hidden form

Adding `t=2025` to the URL did nothing: the site ignores it. Past seasons are chosen through an ASP.NET form, so the scraper sends the same POST request a browser does, carrying the page's `__VIEWSTATE` and `__EVENTTARGET` fields and the values of two dropdowns (season and phase). The first attempt returned the play-offs, the default phase; the phase is now selected by its text, "Liga Regular".

2.5 Validation

1. Team points summed from the box scores against each team's official total.
2. Every game's summed points against its official result.

All checks pass except one documented gap: the 2025/26 box scores of Movistar Estudiantes miss 372 points (12.4 per game), a gap in the source itself, confirmed on two separate pages of the site.

2.6 Season totals, heights and positions

Box scores are summed per player link. Because the link contains the team, each row is a player-team stint, which is also the paper's unit of analysis; eleven players had two stints in 2025/26. Heights come from the player pages: 27 were missing and were completed by hand from public sources, recorded in `manual_heights.csv` with their source. Positions arrived in 15 spellings (including "Esc" and "A-piv") and were standardised to PG, SG, SF, PF and C with `unicodedata` and a mapping dictionary.

One lesson in reproducibility came from a simple error: running a cleaning cell twice (`x = f(x)`) cleaned the data twice and broke it (26 columns where 20 were expected). From then on, raw and clean tables are kept separate (`_raw` tables, a `height_feb` column created once), so every cell gives the same result however many times it runs.

2.7 Team tables and possessions

The official team tables (regular season only) give each team's totals. Possessions are estimated from them:

$$\text{Poss} = \text{FGA} - \text{OREB} + \text{TOV} + 0.44 \times \text{FTA}$$

The notebook saves eight files for the next stage: box scores, season totals, games and team tables for each season.

3. The model (notebook 02_model)

The model notebook turns season totals into the paper's 15 features, fits Archetypal Analysis, tests how far its results can be trusted, and projects them onto 2026/27.

3.1 Team and opponent context

Rates such as "share of available rebounds" need to know what happened on court while the player was there, so every player row is joined to her team's context. Team totals (minutes, possessions, field goals, rebounds, turnovers, free throws) come from the official team tables; opponent totals (field-goal and two-point attempts, rebounds, turnovers, free throws) are summed from the box scores by opponent, and opponent possessions follow the same formula as in Section 2.7.

3.2 The 15 features

All rates follow the Basketball-Reference definitions, which scale a player's events by her share of the team's playing time:

$$\text{on_court} = \frac{\text{MIN}}{\text{Team MIN}/5}$$

Where a ratio is 0/0 (a player who never shot a three, for instance), it is set to 0. A built-in check: usage rates of all players must average 20%, since five players share every possession. In the sample the mean is 20.00%.

3.3 The sample

As in the paper, only stints with at least 300 minutes enter the model: 142 stints in 2025/26. For the 2026/27 projection, with only seven rounds played, the threshold is 70 minutes.

3.4 First run, and why shooting percentages are shrunk

The first fit used raw shooting percentages, and some archetypes were driven by players with extreme values on very few attempts: a player who goes 5 for 8 from three looks like an elite shooter on almost no evidence. The second run, kept below the first in the notebook, shrinks each percentage towards the

Feature	Definition
OREB%	OREB / (on_court × (Team OREB + Opp DREB))
DREB%	DREB / (on_court × (Team DREB + Opp OREB))
AST%	AST / (on_court × Team FG − FG)
TOV%	TOV / (FGA + 0.44 × FTA + TOV)
STL%	STL / (on_court × Opp possessions)
BLK%	BLK / (on_court × Opp 2-point attempts)
USG%	(FGA + 0.44 × FTA + TOV) / (on_court × (Team FGA + 0.44 × Team FTA + Team TOV))
PTS/100, FGA/100	Points and field-goal attempts per 100 of the player's possessions (on_court × team possessions)
FT rate	FTA / FGA
3PA share	3PA / FGA
2P%, 3P%, FT%	Shooting percentages, shrunk (Section 3.4)
Height	Centimetres

league average by adding imaginary attempts at that average:

$$3\text{P\%}_{\text{shrunk}} = \frac{3\text{PM} + 50 \times 0.321}{3\text{PA} + 50}$$

The league averages come from the 2025/26 sample (47.3% from two, 32.1% from three, 74.4% from the line), with 50 imaginary attempts for twos and threes and 30 for free throws. A player with 300 three-point attempts barely moves; one with 8 moves most of the way to the average.

3.5 Scaling

Features are standardised to z-scores with a scaler fitted on 2025/26 only. The same scaler, unchanged, is applied to 2026/27, so both seasons are measured on the same ruler.

3.6 Archetypal Analysis against K-Means

Archetypal Analysis (AA) finds K extreme profiles, the archetypes, and writes every player as a mixture of them, with non-negative weights that sum to one:

$$x_i \approx \sum_{k=1}^{K} w_{ik}\, a_k, \qquad w_{ik} \geq 0, \qquad \sum_{k=1}^{K} w_{ik} = 1$$

Each archetype is itself a mixture of real players, so it sits at the edge of the data; a K-Means centroid sits at the centre of its cluster and gives each player a single label. Both were fitted for K = 2 to 12 (with the Python package `archetypes` and scikit-learn), measuring the share of variation each explains (1 − residual sum of squares / total). AA explains more at every K: 68.9% against 49.6% at K = 8 (Figure 1).

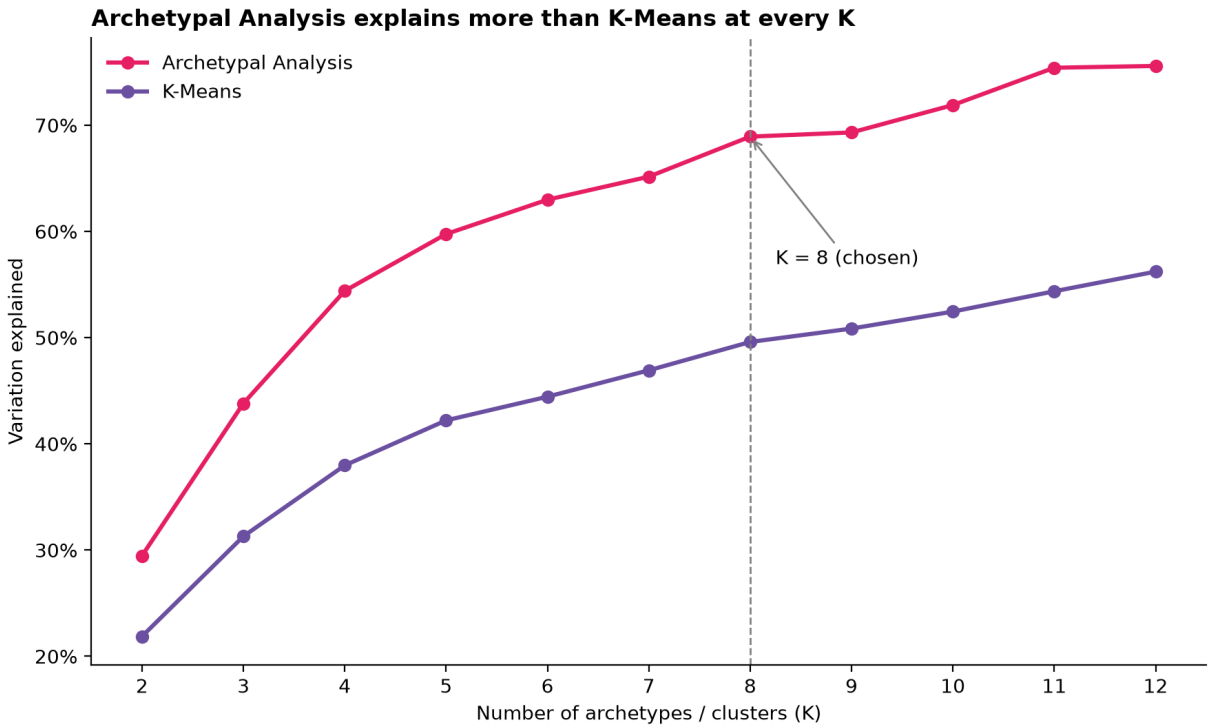


Figure 1 — Variation explained by Archetypal Analysis and K-Means, K = 2 to 12.

3.7 Choosing K

The curve has no single sharp elbow. The gain is almost flat from 8 to 9 archetypes (+0.4 points), but it rises again at 10 and 11, so K = 9 was considered explicitly. K = 8 was kept on four grounds together: the 8-to-9 plateau, the basketball interpretability of the eight archetypes, their stability with a sample of 142 stints (more archetypes split it into smaller, less reproducible groups), and direct comparability with the paper's eight archetypes. Section 5.2 adds the paper's own diagnostic for K.

3.8 The final model and its names

AA can land on different solutions from different starting points, so each K was fitted from 20 random starts and the best fit kept, as K-Means does with `n_init`. The final model (start 0) explains 69.5% of the variation. The archetypes had first been named on an exploratory fit (start 42), by reading each one's standardised profile and the players with the highest weights; the Hungarian algorithm then matched the final archetypes to those names one to one.

3.9 Stability across starts

To know which archetypes are real and which are artefacts of one fit, the model was refitted 20 times with different seeds, for K = 6, 7 and 8. Each refit's archetypes were matched to the final ones with the Hungarian algorithm on their distances (`linear_sum_assignment` on `cdist`), and an archetype counts as found again when its match lies within a distance of 1.5.

A first reading of this table, "perimeter archetypes are stable, interior ones form a continuum", turned out to be wrong: Glass Cleaner is interior and solid, Scoring Wing is perimeter and tentative. What the table follows is usage. Across the eight archetypes, stability and usage correlate at −0.81 (Figure 2): **role players come in types, stars don't.** There is a typical way to be a spot-up shooter or a pass-first organiser; each player who carries the offence does it her own way. Glass Cleaner is the exception, stable despite average usage, probably because it is defined by physical traits rather than by what the player does with the ball. With eight points, the correlation illustrates the pattern rather than testing it. In practice, solid archetypes can be used as labels, while tentative ones are better read as directions within a player's mix.

Stability	Archetype	Found again	Typical usage
Solid	Spot-Up Shooter	89%	9.5%
Solid	Floor General	74%	14.5%
Solid	Glass Cleaner	74%	22.0%
Moderate	Ball-Hawk Guard	58%	24.6%
Moderate	Rim Protector	53%	18.5%
Tentative	Combo Creator	42%	24.8%
Tentative	Interior Star	42%	23.8%
Tentative	Scoring Wing	26%	25.1%

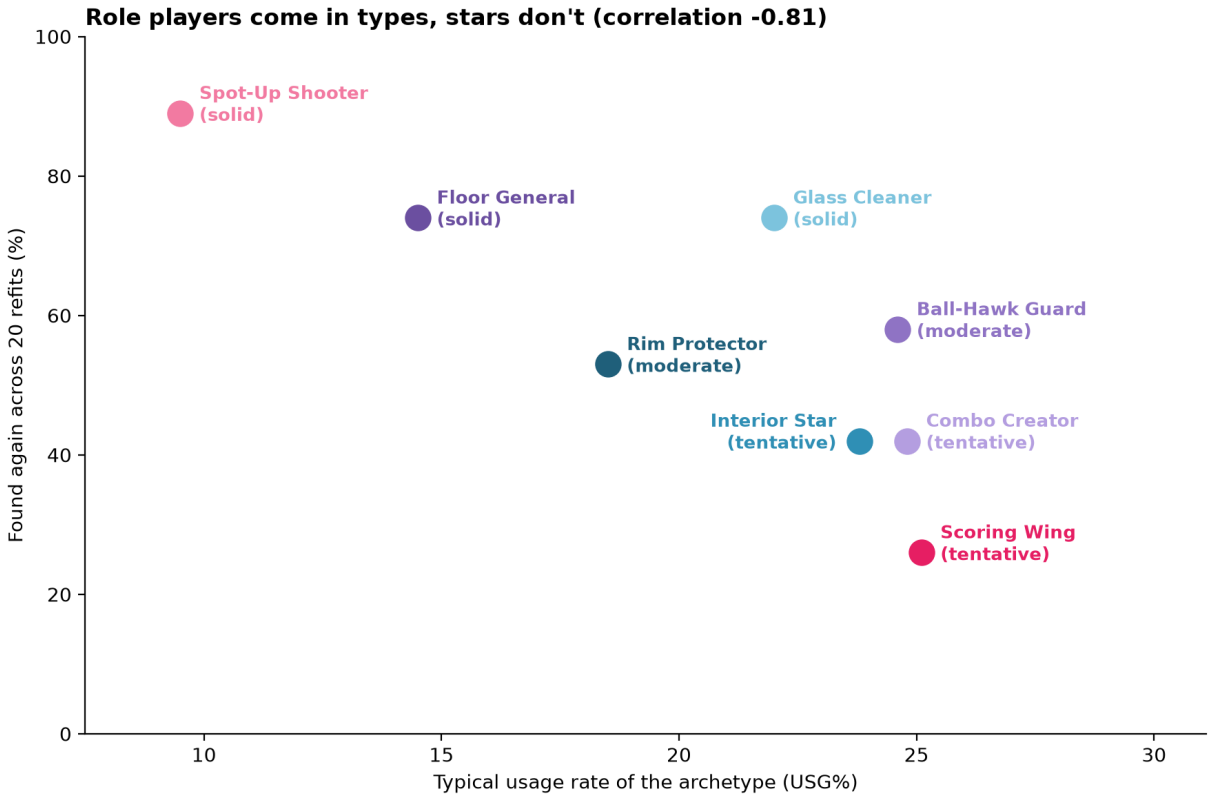


Figure 2 — Archetype stability across 20 refits against typical usage.

3.10 Archetypes against positions

Positions capture roles only in part, and most players are genuine mixtures. 59% of players have no archetype above 50% of their mix. Grouping the archetypes into the family each position would expect (guards, wings, bigs) and measuring how much of each position's mix falls in its own family gives centres 81%, point guards 66%, shooting guards 53%, power forwards 52% and small forwards 49%. Two players show the gap clearly: Nia Coffey, listed as a small forward, is 97% Rim Protector, and Chloe Bibby, listed as a power forward, is 80% Scoring Wing.

3.11 Similar players with KNN

For each player, cosine K-nearest neighbours finds the most similar profiles, in two spaces: the 15 standardised features, and the eight archetype weights. The player's own other stint is excluded, so a player who changed clubs does not find herself. Iyana Martín (Perfumerías Avenida) is 50% Scoring Wing, 38% Ball-Hawk Guard, 13% Combo Creator and 0% Floor General: a point guard who scores, creates and steals but does not organise. Her nearest neighbour is Imani Tate, and around both sits a family of six or seven scoring, pressuring guards. Claudia Contell, who played for two clubs in 2025/26, shows a different profile in each stint, a first sign that roles depend on context.

3.12 Projecting 2026/27 with the frozen model

The 2026/27 players (70+ minutes, seven rounds) were standardised with the 2025/26 scaler and projected with the 2025/26 model, with no refitting. Of the 77 players present in both samples, 55% keep their main archetype. Some changes are plausible (Moira Joiner and Madison Conner move from Scoring Wing to Combo Creator after changing clubs); others are not (Veronika Voráčková, same team, from Floor General to Rim Protector). Seven games are a small sample, so before reading any change, the noise had to be measured.

3.13 How much is noise? A baseline

The same comparison was run where no real change is possible: 2025/26 rounds 1 to 7 against the full 2025/26 season, for the same players on the same teams. The first attempt failed on 11 missing heights: players with 70+ minutes in the first seven rounds who never reached 300 minutes, so their heights had never been completed. They had no full-season mix to compare with anyway, so the baseline keeps the 118 stints present in both versions. Change is measured as the share of the mix that moves: 0.5 × the sum of absolute weight differences (0 = identical, 1 = completely different).

Comparison	Players	Median change	90th percentile	Keep main archetype
Noise only: rounds 1–7 vs full season	118	0.25	0.50	61%
Real: 2025/26 vs 2026/27	77	0.42	—	55%

Noise alone changes the main archetype of 39% of players, so most 2026/27 archetype switches are within what chance produces. The mix itself, however, moves more between seasons than within one (median 0.42 vs 0.25). Using the 90th percentile of noise (0.50) as a threshold, 23 of 77 players (30%) change more than chance would explain, three times the 10% expected (Figure 3). Only real change reaches the 0.75–1.0 range: Moira Joiner (0.97), Alessandra Orsili (0.94), Veronika Voráčková (0.89), Itziar Ariztimuño (0.87), Carla Osma (0.80) and Madison Conner (0.80). Because rounds 1–7 are part of the full season, this baseline slightly underestimates noise, so the conclusions are conservative.

3.14 Club changes and league drift

A first count found 42 of 77 players changing clubs, which was too many: several teams had only changed name. Listing the team names of both seasons identified five renames (four sponsor changes and an

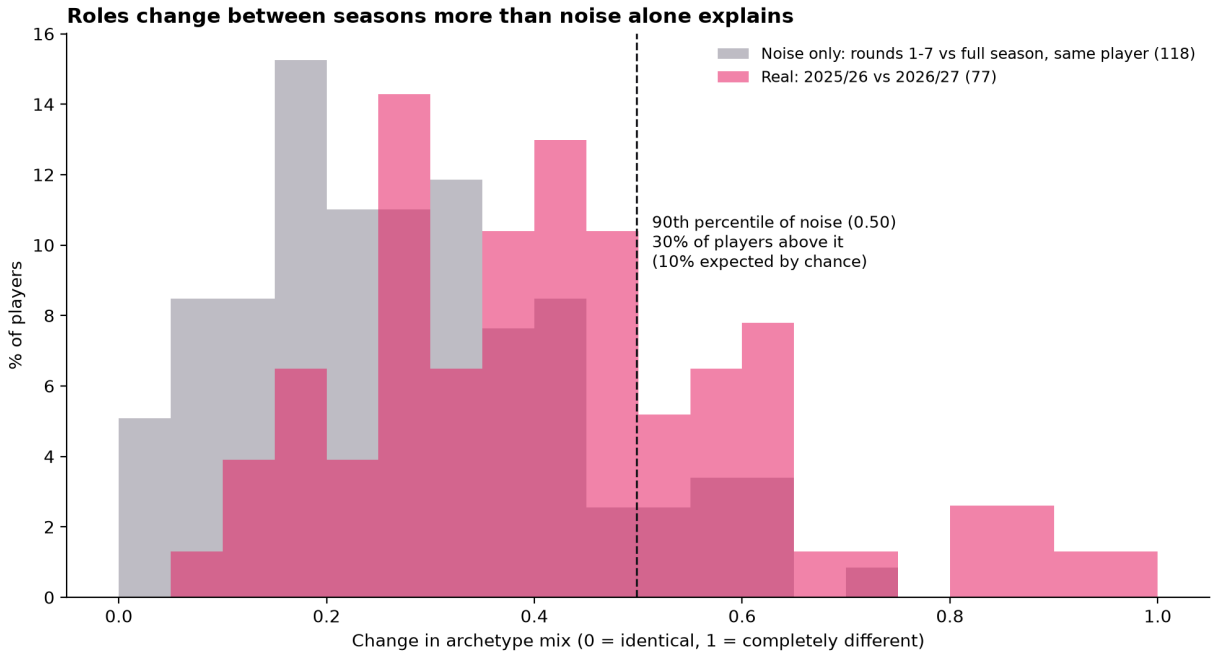


Figure 3 — Change in archetype mix: noise only against real change between seasons.

accent, "Leganes" to "Leganés"), alongside two relegated and two promoted teams. With the renames mapped, 35 players changed clubs. They exceed the noise threshold slightly more often (34% vs 26%) and change more (median 0.46 vs 0.38), consistent with a new team bringing a new role, but the groups are small (12 vs 11 players), so this is suggestive. Players who stayed also change far more often than chance: roles move between seasons even without a transfer. Finally, the average weight of each archetype moves by at most 0.03 between seasons, so there is no league-wide drift.

The practical rule for scouts: read early-season archetypes as a mix with uncertainty, not as a label, and check them again after more rounds.

4. Results (notebook 03_results)

The results notebook never refits anything: it reads the model's saved outputs and turns them into archetype portraits, team analysis, player cards and an interactive map.

4.1 From model to results

Each notebook has its own memory, so 02_model ends by saving its outputs: every player's archetype mix for 2025/26 and 2026/27, the season-to-season comparison, the standardised features and the archetype profiles. A check confirms that the saved profiles belong to the final model: rebuilding every player as her weights times the eight profiles recovers 69.5% of the variation, exactly the final model's figure.

4.2 Portraits of the eight archetypes

The model finds five perimeter archetypes and three interior ones, each with a clear basketball reading and, in most cases, a real player who embodies it.

Three visual designs were tried before settling on one. A full heatmap of the eight archetypes by 15 features, coloured by distance from the league average and labelled with actual values, holds everything but asks the reader to scan 120 cells (Figure 4). Small bar charts of each archetype's five most distinctive stats were easier to read one by one but showed different stats in each panel, so archetypes could not be compared. The final design gives each archetype the same percentile radar as the player cards, with the same 15 axes in the same order (Figure 5): comparing archetypes becomes comparing shapes. Colour in the heatmap means more or less than average, not better or worse: a red turnover rate means more

Archetype	Defining values	Purest player
Floor General	26.0% assist rate, 30.8% turnover rate, 10.3 points per 100, 14.5% usage	Lucía Togores (99%)
Ball-Hawk Guard	6.1% steal rate, 26.2% assist rate, 24.6% usage, 25% from three, 170 cm	Claudia Soriano (100%)
Combo Creator	23.7 shots per 100, 24.8% usage, 22.3% assist rate, 8.8% turnover rate, 39% from two	Nikola Dudasova (98%)
Scoring Wing	28.0 points per 100, 39% from three, 60% of shots from three, 25.1% usage	Madison Conner (100%)
Spot-Up Shooter	9.5% usage, 75% of shots from three	Zoe Hernández (100%)
Rim Protector	3.8% block rate, 22.1% defensive rebound rate, 190 cm	Nia Coffey (97%)
Interior Star	29.1 points per 100, 56% from two, 13.8% offensive rebound rate, 193 cm	Aminata Gueye (92%)
Glass Cleaner	10.9% offensive rebound rate, 0.53 free-throw rate, 51% at the line, 193 cm	Lauren Ebo (100%)

turnovers. In the radars, turnovers are inverted as ball security, so further out is always better.

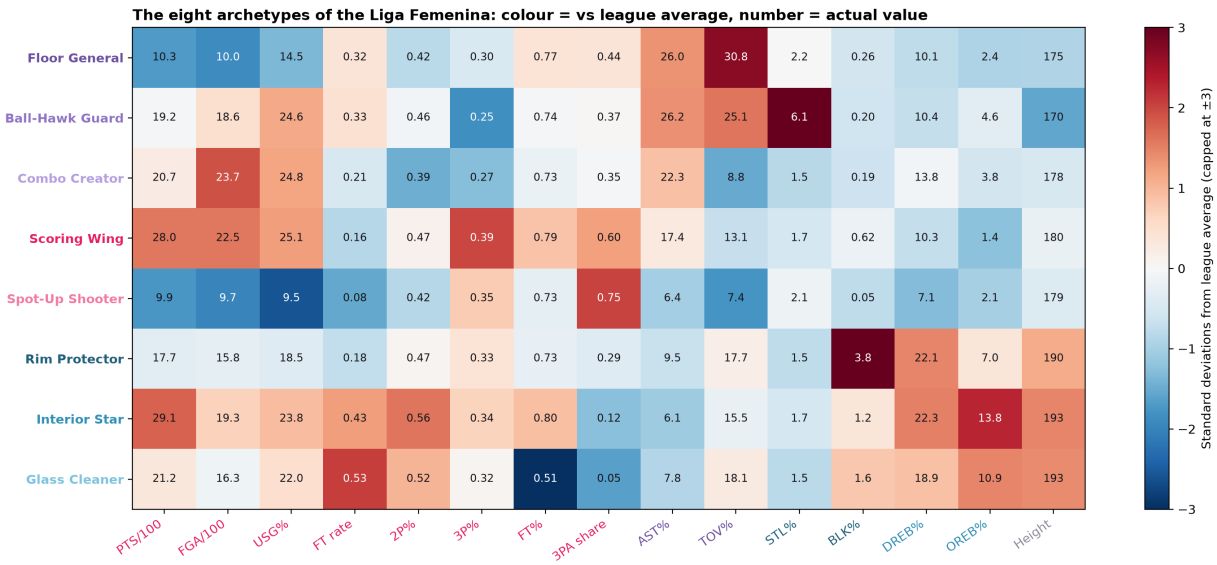


Figure 4 — The eight archetypes: colour = standard deviations from the league average, number = actual value.

4.3 Glossary

Each row of Figure 4 is a pure, imaginary player who is 100% one archetype; real players are mixtures and sit between the rows. The Floor General's 30.8% turnover rate is a share of few plays: she uses only

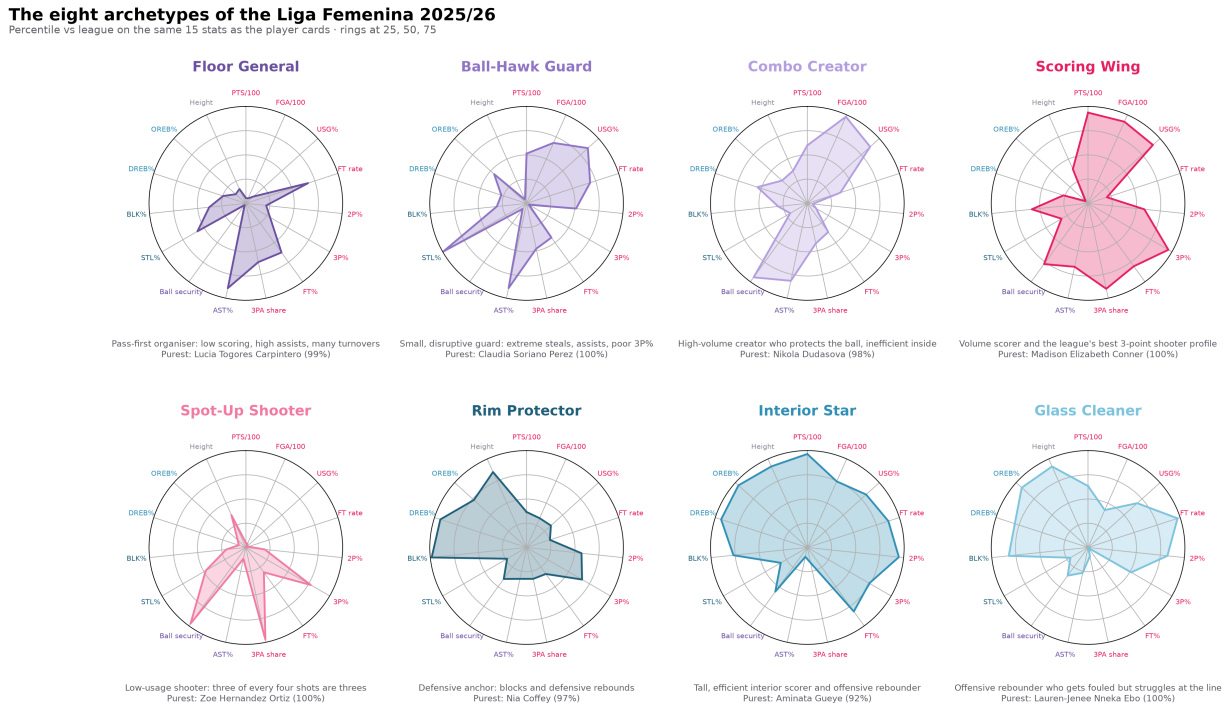


Figure 5 — The eight archetypes as percentile radars, with their purest players.

Feature	Meaning	Example from Figure 4
PTS/100, FGA/100	Points and shots per 100 possessions played	Floor General 10.3: about 10 points per 100 possessions
USG%	Share of team possessions used while on court	Spot-Up 9.5: one possession in ten ends with her (average 20)
FT rate	Free-throw attempts per field-goal attempt	Glass Cleaner 0.53: 53 free throws per 100 shots
2P%, 3P%, FT%	Shooting percentages, shrunk	Scoring Wing 0.39: 39% from three
3PA share	Share of shots that are threes	Spot-Up 0.75: three shots in four
AST%	Share of teammates' baskets she assisted while on court	Floor General 26.0: one in four
TOV%	Turnovers per 100 plays she uses	Floor General 30.8
STL%	Share of opponent possessions ending in her steal	Ball-Hawk 6.1
BLK%	Share of opponent two-point attempts she blocks	Rim Protector 3.8
DREB%, OREB%	Share of available rebounds she collects	Interior Star OREB 13.8
Height	Centimetres	Ball-Hawk 170

14.5% of possessions, so each turnover weighs heavily.

4.4 Case study: Valencia B.C., the 2025/26 champion

The champion built from the rim out, without a classic organiser. A team's archetype mix is the minute-weighted average of its players' mixes: a player with 1,000 minutes weighs more than one with 300. Only players in the sample (300+ minutes) enter, so this describes the main rotation.

Archetype	Valencia	League	Difference
Rim Protector	0.207	0.089	+0.119
Scoring Wing	0.176	0.145	+0.031
Ball-Hawk Guard	0.126	0.103	+0.023
Spot-Up Shooter	0.146	0.133	+0.013
Interior Star	0.180	0.171	+0.009
Combo Creator	0.091	0.125	−0.035
Glass Cleaner	0.016	0.084	−0.069
Floor General	0.058	0.150	−0.092

- **More than twice the league's Rim Protector share**, carried by María Pérez Araujo (0.52), Raquel Carrera (0.40) and Nia Coffey (0.97).
- **Almost no classic Floor General.** Its three point guards, Leticia Romero, Hind Ben Abdelkader and Ivon Anderson, all profile mainly as Scoring Wings: creation is shared rather than run by one organiser.
- **Almost no Glass Cleaner.** Rebounding comes from rim protectors and interior stars such as Awa Fam Thiam (47% Interior Star) and Kayla Alexander (82%), who do more than rebound.
- **A deep rotation.** Eleven players reach 300 minutes and none exceeds 650, so the profile does not rest on a single player.

4.5 Team fingerprints for all 16 teams

Every team's identity comes down to one or two archetypes, worth roughly one rotation player. Two views show it. The fingerprint heatmap gives each team's mix minus the league average, with a diverging scale centred on zero (Figure 6); the largest deviations are about ±0.12. The composition chart gives each rotation's absolute mix, with the league average on top as a reference (Figure 7). All 16 rotations contain all eight archetypes and resemble the league average more than they differ from it.

Team	Distinctive block	Team share	League share
Spar Girona	Interior Star	28%	17%
Innova-TSN Leganés	Combo Creator	27%	13%
Movistar Estudiantes	Floor General	23%	15%
Baxi Ferrol	Scoring Wing	23%	15%
Valencia B.C.	Rim Protector	21%	9%
Club Joventut Badalona	Glass Cleaner	19%	8%

Kutxabank Araski shows the largest gap in the league (Interior Star −0.12), and Perfumerías Avenida stands out for Ball-Hawk Guards (+0.09). Teams were sorted by their interior share, expecting a gradient

from perimeter to interior teams; none appeared. Team identities are combinations of archetypes, not points on a single perimeter-to-interior scale.

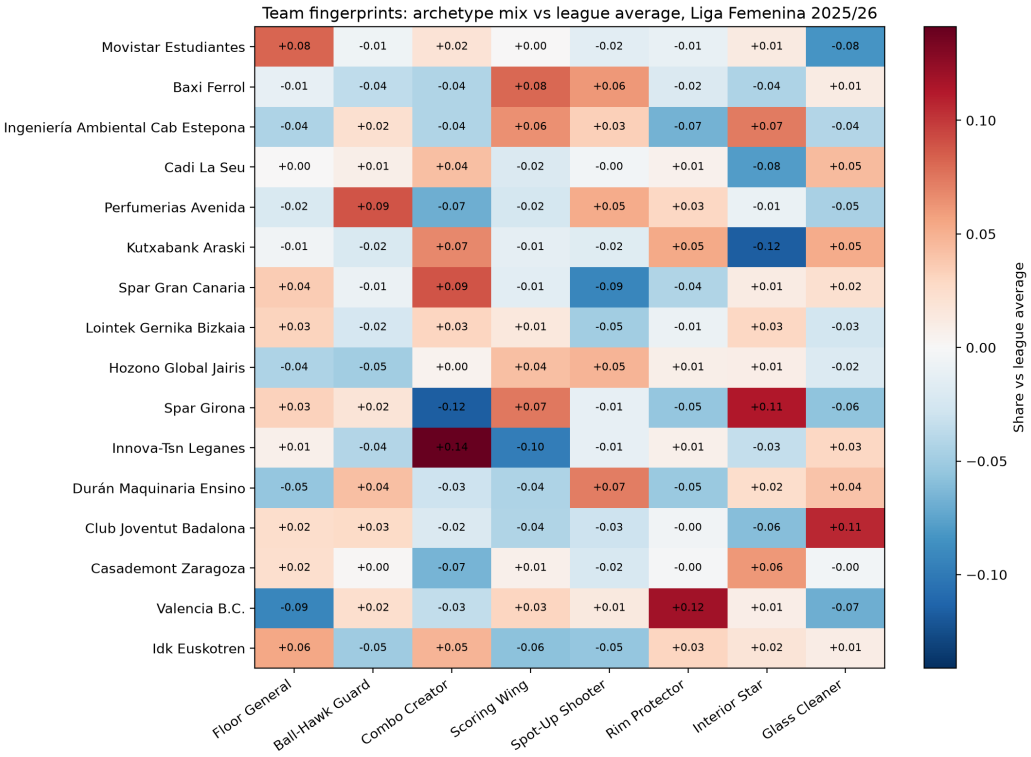


Figure 6 — Team fingerprints: each team's archetype mix minus the league average.

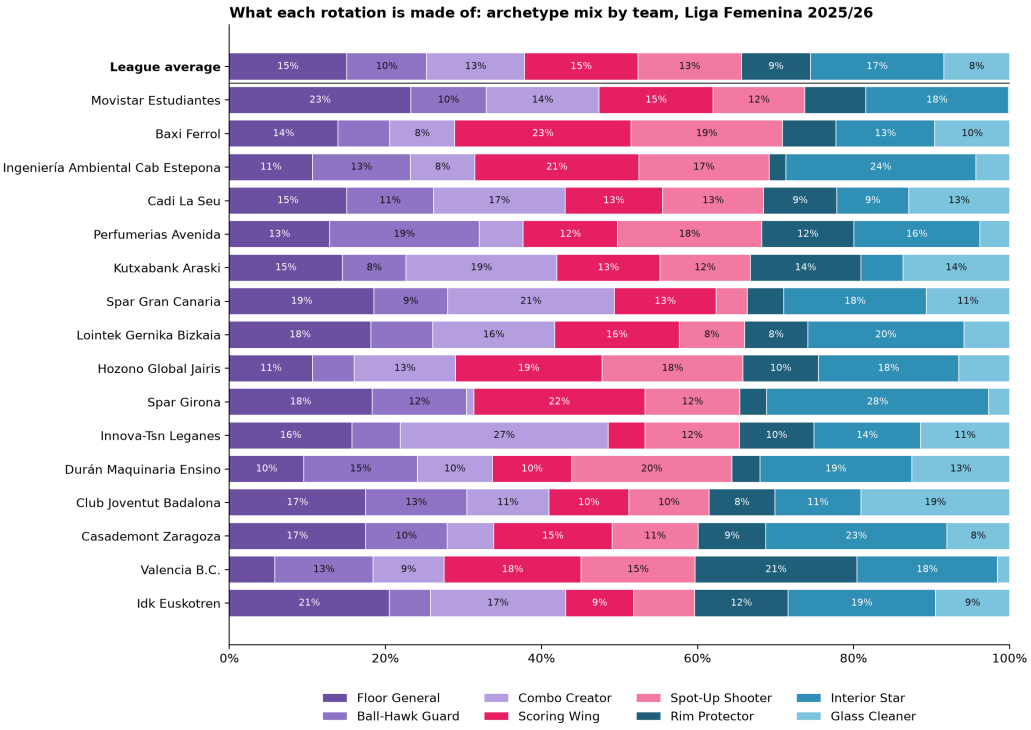


Figure 7 — What each rotation is made of: archetype mix by team, with the league average on top.

4.6 Player cards in a loop

Every one of the 142 stints gets a card, produced by one function called in a loop, so all cards are built the same way. Each card has three panels: the archetype mix as bars in the archetype colours, a percentile radar of the 15 features (built with `melt` into one row per player and feature, percentiles within the sample, turnovers inverted as ball security), and a sheet with team, position, height, minutes, main archetype and the three most similar players by cosine KNN on the standardised features (Figure 8).

The first version worked but its radar followed the column order, so its shape said little. Version 2 groups the axes by theme (scoring, creation, defence, rebounding, size), colours the labels by group, starts at the top, writes names as "First Last", adds each similar player's team, and skips a player's other stint so no one appears twice. Every card is saved as a PNG and closed inside the loop, so 142 figures never sit in memory at once.

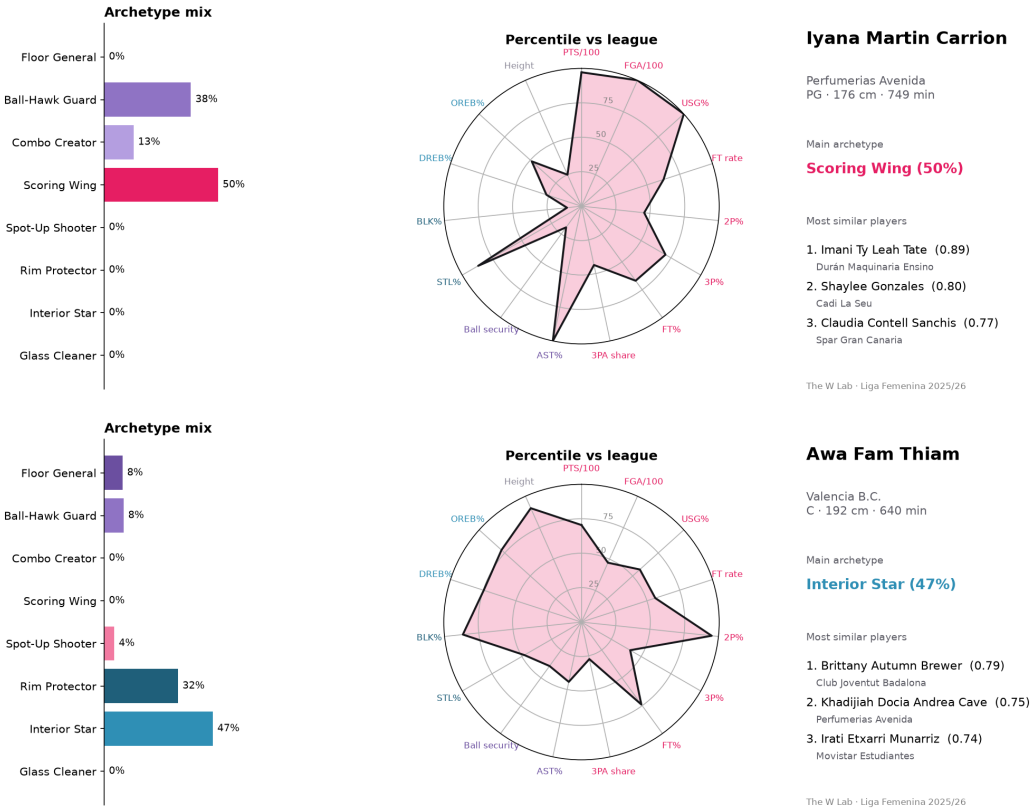


Figure 8 — Player cards for Iyana Martín and Awa Fam Thiam.

Reading the cards. Iyana Martín (Perfumerías Avenida, point guard, 176 cm, 749 minutes) is 50% Scoring Wing, 38% Ball-Hawk Guard and 13% Combo Creator, with nothing of the Floor General. Her radar explains the mix: she sits near the top of the league in usage, shots and points per 100 possessions, and her assist and steal rates are also among the highest, but she is in the lowest quarter for ball security, height and rebounding, and below average in her share of threes. She is a high-volume creator who scores, passes and pressures, and who pays for her volume in turnovers. Her closest profiles, Imani Tate (Durán Maquinaria Ensino, similarity 0.89), Shaylee Gonzales (Cadí La Seu, 0.80) and Claudia Contell (Spar Gran Canaria, 0.77), are all scoring, pressuring guards: the family of profiles first seen in Section 3.11.

Awa Fam Thiam (Valencia, centre, 192 cm, 640 minutes) is the interior counterpart: 47% Interior Star and 32% Rim Protector. Her radar peaks in two-point percentage, blocks, height and rebounding at both ends, with average usage and almost no assists or threes: an efficient finisher who also protects the rim, the profile Valencia's rotation is built around (Section 4.4). Her closest profiles are Brittany Brewer (Joventut, 0.79), Khadijiah Cave (Avenida, 0.75) and Irati Etxarri (Estudiantes, 0.74). For a scout, each card answers two questions at once: what kind of player is this, and who else in the league plays like her.

4.7 An interactive map of the league

Principal component analysis projects every player from 15 dimensions onto the two directions that keep most variation, so similar players sit close together. The loadings name the axes:

Axis	Variation kept	Negative end	Positive end
PC1: perimeter to interior	30%	3PA share, AST%, STL%	OREB%, height, PTS/100
PC2: role player to go-to scorer	17%	TOV%, OREB%, BLK%	FGA/100, USG%, PTS/100

An early label error was caught here: the loadings were first printed with "Ball security" for the turnover rate, but the inversion applies only to the radar percentiles, so on the map the variable is the raw TOV%.

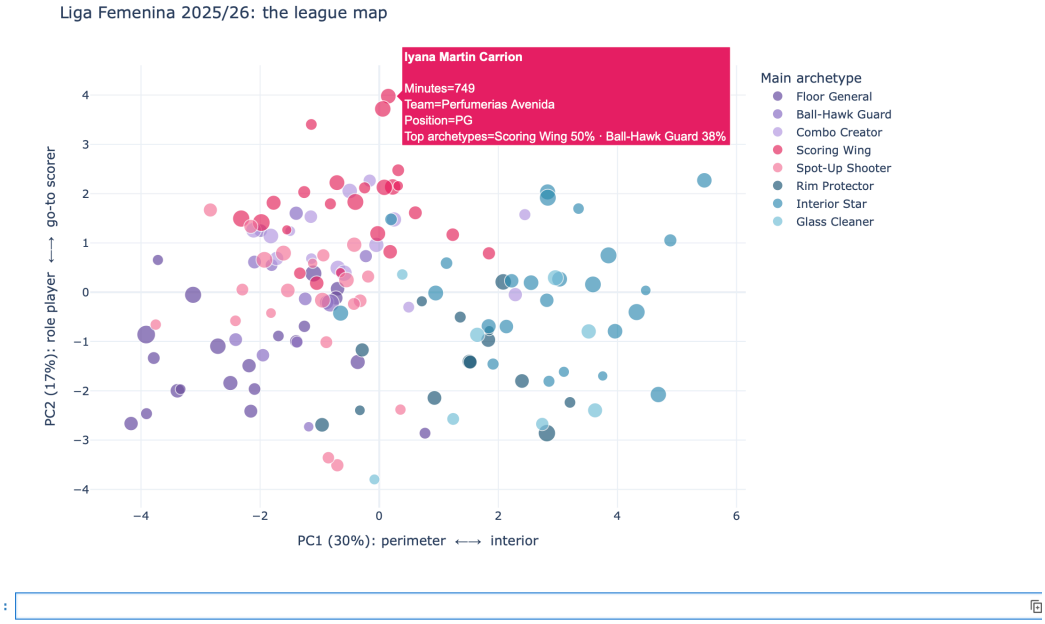


Figure 9 — The interactive league map, 2025/26 (hover shows a player's mix).

The map is built with Plotly: one point per player, coloured by main archetype with the same colours as the cards, sized by minutes, and showing name, team, position and top two archetypes on hover. It is saved as a standalone HTML page. A second version adds 2026/27 with a season slider. The 2025/26 scaler was rebuilt from the same 142 players and checked against the saved z-scores; the 2026/27 players were then projected with `pca.transform`, never refitted, and a player present in both seasons slides across the map. Players the model had never seen fall in the expected regions. The 2026/27 cloud looked wider, but measured it is only slightly so (standard deviation on PC2 1.74 against 1.61, about 8% more; on PC1 2.05 against 2.13): a handful of extreme early-season players drive the impression. With 47% of the variation, the map is a guide for exploring; the eight archetypes, at 69.5%, remain the analysis.

5. Comparison with the paper

Perimeter roles in the Liga Femenina match the paper's men's archetypes closely; the interior is organised differently, and one archetype has no men's counterpart. Because both studies use the same recipe, these differences reflect the leagues, not the method.

	Paper (men)	This study (women)
Data	7 European domestic leagues incl. the Spanish ACB, 2014–2025	Liga Femenina Endesa, 2025/26
Features, filter, scaling	15 box-score features, 300 minutes, z-scores	Same
Unit, model, K	Player-season-team, Archetypal Analysis, 8	Same
Sample	Thousands of player-seasons	142 stints

5.1 Which archetypes match

The paper publishes its archetypes as verbal descriptions, not numbers. Each description was encoded as signs over the 15 features (+1 for "high", −1 for "low", 0 if not mentioned), and each pair of archetypes received an agreement score: how many standard deviations our archetype lies, on average, in the direction the paper describes (Figure 10). To check that the encoding does not drive the result, the matching was repeated 34 times, each time dropping one descriptor.

Paper archetype (men)	Best match in the Liga Femenina	Agreement	Robustness
Traditional Playmaker	Floor General	+1.66	100%
3PT Specialist	Spot-Up Shooter	+1.44	94%
High Usage Guard	Scoring Wing	+1.04	97%
High Usage Forward	Interior Star	+1.23	100%
Defensive Specialist	Rim Protector	+0.91	94%
Traditional Center	Glass Cleaner	+1.38	91%
Backup Big	Glass Cleaner	+0.85	91%
Role Guard	Ball-Hawk Guard	+1.35	97% (two descriptors only)

Four findings follow:

1. **Role archetypes are shared across men's and women's basketball.** The two closest pairs, Floor General and Spot-Up Shooter, are also our two most stable archetypes (Section 3.9).
2. **The interior is organised differently.** Both men's centre types point to our Glass Cleaner; our Interior Star matches the men's High Usage Forward, not a centre; and our Rim Protector matches the men's low-usage Defensive Specialist, so rim protection is a specialised role rather than the classic centre's job.
3. **High-usage roles split.** The men's High Usage Guard agrees with three of our archetypes (Scoring Wing +1.0, Ball-Hawk Guard +0.8, Combo Creator +0.8), and the 3PT Specialist agrees equally with Spot-Up Shooter and Scoring Wing: shooting comes in an off-ball and an on-ball version. These are the archetypes that are least stable within our own league.

4. **Combo Creator has no men's counterpart.** It is the best match of none of the paper's archetypes: a candidate archetype specific to the Liga Femenina.

Under the one-to-one constraint, the perimeter pairs are robust (88–100% of variants) while the interior pairs change in about half of them, because the men's interior archetypes compete for the same players of ours.

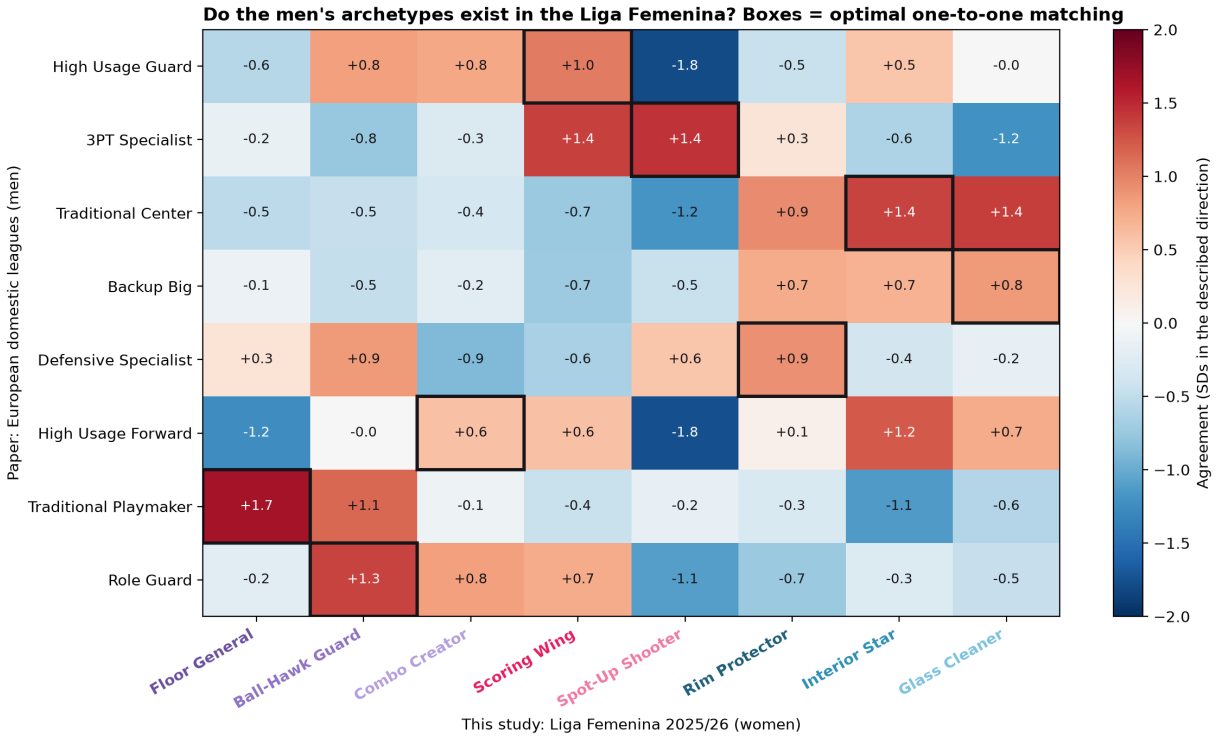


Figure 10 — Agreement between the paper's archetypes and ours; boxes mark the optimal one-to-one matching.

5.2 Does the method behave the same way?

Yes, at a coarse level. The paper chose K = 8 because its intra-archetype variance (how similar the players within each archetype are) reached a minimum there. In the Liga Femenina the same measure falls steadily from 0.621 at K = 4 to 0.549 at K = 9, with no minimum (Figure 11). Up to K = 6 our curve is almost identical to the paper's European one (0.607 vs 0.607 at K = 5); beyond that it takes the shape of the paper's NBA curve, where the paper also kept K = 8. All three curves lie between 0.52 and 0.63. The paper does not define its formula, so the comparison of levels is indicative.

5.3 Real players at the corners

The paper uses Archetypoid Analysis so that each archetype is a real player. In the Liga Femenina, plain Archetypal Analysis gets close to that on its own: six of the eight archetypes have a real player almost on top of them, against a typical player-to-archetype distance of 3.63. Madison Conner is the Scoring Wing (distance 0.00); Claudia Soriano (0.24), Nikola Dudasova (0.30), Lauren Ebo (0.39), Zoe Hernández (0.51) and Lucía Togores (1.09) anchor their archetypes in the same way. The two exceptions are interior: no real player is close to the pure Rim Protector (2.15) or Interior Star (1.41), the same archetypes whose match to the paper is least clear.

5.4 Players are mixtures

The paper's central claim holds, and more strongly than in its own examples. The typical Liga Femenina player has no majority archetype: most main-archetype weights lie between 35% and 55%, below the paper's examples of Antetokounmpo (53.5%) and Osetkowski (57%). Only 6% of players exceed the 90% memberships the paper attributes to Gaussian mixture models (Figure 12). The 12% above Markus

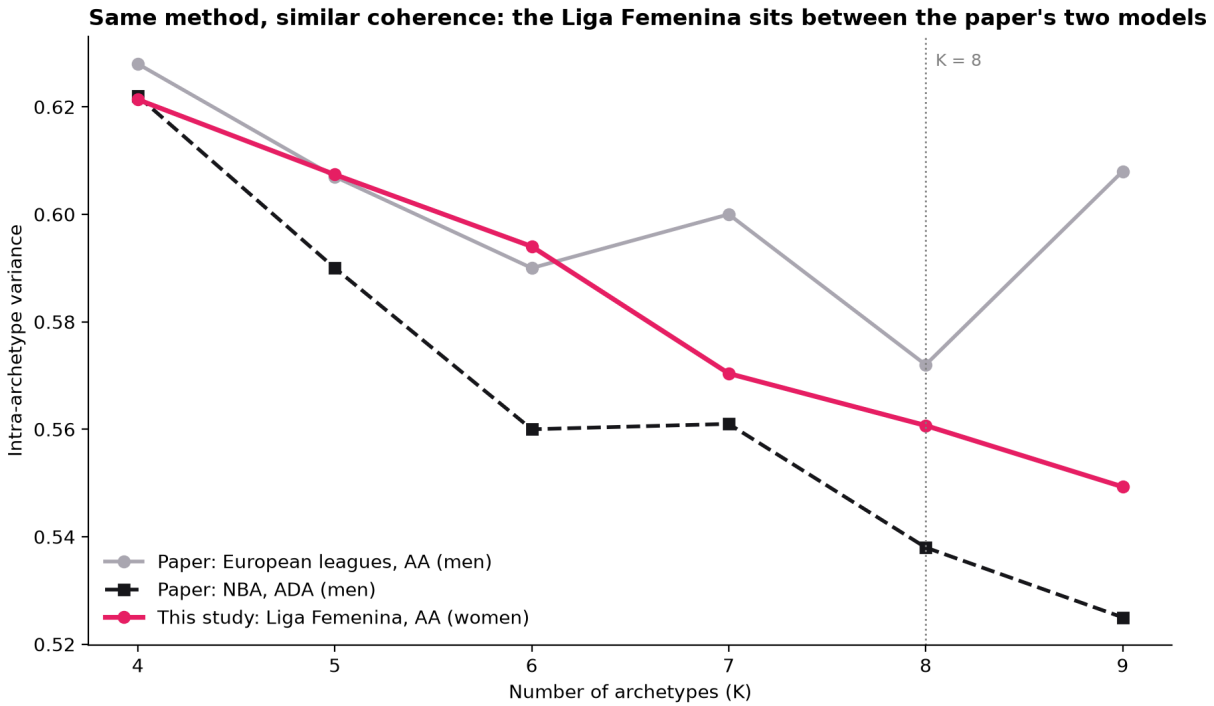


Figure 11 — Intra-archetype variance by K: this study against the paper's two models.

Howard's 72%, described in the paper as one of the highest concentrations in the EuroLeague, are mostly the players who anchor the archetypes (Section 5.3).

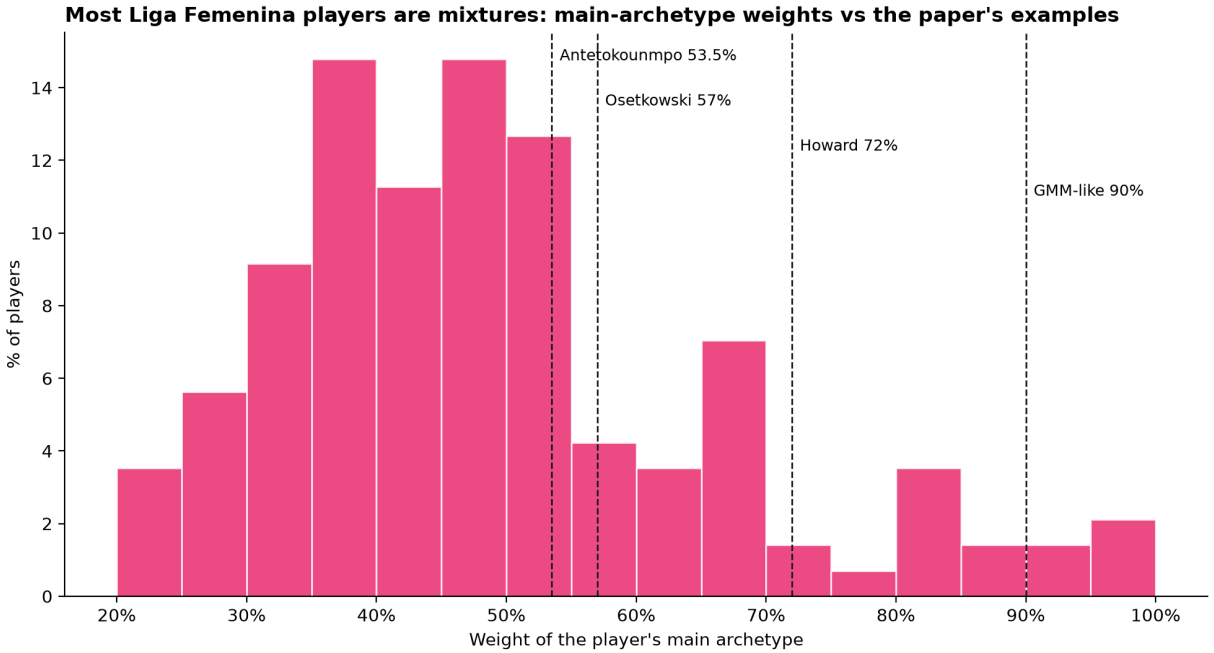


Figure 12 — Distribution of main-archetype weights against the paper's examples.

5.5 What this study adds

- A women's league, testing the "any league" claim outside men's basketball.
- The stability of each archetype across refits.
- A projection with the frozen model to a new season, with a baseline for early-season noise (Section 3.13).

- Shrinkage of shooting percentages for small samples.
- Measurements of two claims the paper makes in words: that positions miss roles (Section 3.10) and that roles change with context (Section 3.14).

The paper, in turn, has what this study lacks: lineup-level analysis with on/off net ratings, and Archetypoid Analysis proper.

6. Conclusions

The "any league" claim of *Scouting Anyone* survives its first test outside men's basketball, with one refinement: the method finds the types that exist, and in the Liga Femenina those types are the role players. Applied with the paper's exact recipe to 142 stints of the 2025/26 season, Archetypal Analysis produces eight interpretable archetypes that explain 69.5% of the variation, 20 points more than K-Means (49.6%). The archetypes that match the men's model best are the same ones that are most stable within our league, and the ones that differ are the same ones the model cannot pin down.

Question	Short answer	Key numbers
Do the paper's archetypes exist in a women's league?	Yes for role players; the interior and the creators are organised differently	3 robust one-to-one matches (88–100%); Combo Creator matches no men's archetype
Are they stable enough to trust?	The role-player archetypes are; the star archetypes are not	Spot-Up Shooter found again 89%, Scoring Wing 26%; stability vs usage r = −0.81
Do they hold for a new season?	The model holds; early-season labels do not	Noise alone flips 39% of labels; 30% of players change beyond noise vs 10% by chance

6.1 Do the paper's archetypes exist in a women's league?

Yes for the role players, and the agreement is measurable. Encoding the paper's descriptions as signs over the same 15 features and comparing them with our profiles gives three one-to-one matches that survive dropping any single descriptor:

Paper (men)	Liga Femenina (women)	Agreement	Kept in variants
Traditional Playmaker	Floor General	+1.66 SD	100%
3PT Specialist	Spot-Up Shooter	+1.44 SD	91%
High Usage Guard	Scoring Wing	+1.04 SD	88%

The method also behaves the same way. Intra-archetype variance at K = 8 is 0.561 in the Liga Femenina, between the paper's European model (0.572) and its NBA model (0.538); up to K = 6 our curve and the European one are almost identical (0.607 and 0.607 at K = 5).

The differences are structural, not random.

- **The interior is organised differently.** Each men's archetype has a robust favourite among ours (91–100% of variants), but the one-to-one pairs of the interior hold in only 53–56% of variants, because the men's bigs compete for the same players. Both men's centre types point to our Glass Cleaner (+1.38 and +0.85). Our Interior Star, the league's top scorer profile (29.1 points per 100, 56% from two, 13.8% offensive rebound rate), corresponds to the men's High Usage Forward (100% of variants), not to a centre. Our Rim Protector (3.8% block rate, 18.5% usage) corresponds to the men's low-usage Defensive Specialist (94%): in the Liga Femenina, protecting the rim is a specialist's job, not the classic centre's.

- **Creation and shooting split.** The men's High Usage Guard agrees with three of our archetypes at once (Scoring Wing +1.0, Ball-Hawk Guard +0.8, Combo Creator +0.8). The 3PT Specialist agrees equally with the Spot-Up Shooter and the Scoring Wing (+1.4 each): shooting comes in an off-ball version (9.5% usage, 75% of shots from three) and an on-ball version (25.1% usage, 39% from three).
- **Defence comes with the ball.** The league's most extreme stealer, the Ball-Hawk Guard (6.1% steal rate), is also a high-usage playmaker (24.6% usage, 26.2% assist rate), not a low-usage defensive specialist as in the men's model.
- **One archetype is new.** The Combo Creator (23.7 shots per 100, 8.8% turnover rate, 39% from two) is the best match of none of the paper's eight archetypes: a candidate role specific to the Liga Femenina, embodied by Nikola Dudasova (98%).

The paper's central claim holds more strongly here than in its own examples. Most main-archetype weights lie between 35% and 55%, and 59% of players have no archetype above half of their mix, below the paper's examples of Antetokounmpo (53.5%) and Osetkowski (57%). Only 6% exceed the 90% memberships the paper attributes to Gaussian mixtures. Positions capture roles only in part: the share of each position's mix in its own family falls from 81% for centres to 66% for point guards, 53% for shooting guards, 52% for power forwards and 49% for small forwards. Nia Coffey, a small forward, is 97% Rim Protector.

Archetypal Analysis delivers what the paper needed Archetypoid Analysis for. Six of our eight archetypes have a real player almost on top of them, against a typical player-to-archetype distance of 3.63: Madison Conner is the Scoring Wing (distance 0.00), and Claudia Soriano (0.24), Nikola Dudasova (0.30), Lauren Ebo (0.39), Zoe Hernández (0.51) and Lucía Togores (1.09) anchor theirs. Only the two interior archetypes whose match to the paper is weakest, Rim Protector (2.15) and Interior Star (1.41), have no such player.

6.2 Are the archetypes stable enough to trust?

The role-player archetypes are; the star archetypes are not. Across 20 refits from different random starts:

- **Solid:** Spot-Up Shooter 89%, Floor General 74%, Glass Cleaner 74%.
- **Moderate:** Ball-Hawk Guard 58%, Rim Protector 53%.
- **Tentative:** Combo Creator 42%, Interior Star 42%, Scoring Wing 26%.

Stability falls as usage rises ($r = -0.81$ across the eight archetypes): a player who uses 9.5% of possessions plays a recognisable type, while each player who uses 25% does it her own way. The overlap with Section 6.1 is the study's strongest result: the two archetypes that match the paper most closely, Floor General and Spot-Up Shooter, are two of the three most stable, and the archetypes without a clean men's counterpart are among the least stable. **Role players come in types, within a league and across men's and women's basketball; stars don't.**

The number of archetypes is a judgement, not a measurement. Explained variation rises by only 0.4 points from 8 to 9 archetypes but by more at 10 and 11, and intra-archetype variance falls steadily from 0.621 at $K = 4$ to 0.549 at $K = 9$, with no minimum. $K = 8$ rests on the 8-to-9 plateau, interpretability, stability and comparability with the paper, exactly as the paper's own NBA model did.

For practitioners: Spot-Up Shooter, Floor General and Glass Cleaner can be used as labels; Combo Creator, Interior Star and Scoring Wing are better read as directions within a player's mix.

6.3 Do the archetypes hold for a new season?

The model holds; early-season labels do not. Players the model had never seen, from 2026/27, fall in the expected regions of the league map, whose spread grows only slightly (+8% on the scorer axis), and the league's average mix moves by at most 0.03 per archetype: there is no drift.

Individual labels are another matter. Of the 77 players in both seasons, 55% keep their main archetype,

but noise alone, measured on 118 stints by comparing 2025/26 rounds 1–7 with the full season, already flips 39% of labels (61% kept). The mix itself tells a different story:

	Median change in mix	Above the noise threshold (0.50)
Noise only (same season)	0.25	10% by construction
Between seasons	0.42	30% (23 of 77 players)
Players who changed clubs (35)	0.46	34%
Players who stayed (42)	0.38	26%

Roles do change between seasons, three times more often than chance would produce, and only real change reaches the extreme range: Moira Joiner (0.97), Alessandra Orsili (0.94), Veronika Voráčková (0.89), Itziar Ariztimuño (0.87), Carla Osma (0.80) and Madison Conner (0.80). Changing clubs adds a little, consistent with a new team bringing a new role, but the difference rests on 12 against 11 players; roles move even without a transfer. Because the noise baseline slightly underestimates noise, these estimates are conservative.

6.4 What it means for teams

1. **Scouting with public data.** Every one of the 142 stints has a card with its mix, percentiles and three most similar players, all from box scores anyone can download. Iyana Martín (50% Scoring Wing, 38% Ball-Hawk Guard) finds Imani Tate at a cosine similarity of 0.89: the tool the paper promises for leagues with little data works here.
2. **Roster identity.** Rotations resemble the league average far more than they differ from it; identity comes from one block worth about one player (±0.12 at most). Girona carries 28% Interior Star (league 17%), Leganés 27% Combo Creator (13%), Estudiantes 23% Floor General (15%).
3. **The champion's blueprint.** Valencia, the 2025/26 champion, won with 21% Rim Protector (league 9%), 6% Floor General (15%) and 2% Glass Cleaner (8%), spread across eleven players with 300+ minutes and none above 650. Its point guards score rather than organise; its bigs protect the rim and score rather than only rebound.
4. **Early-season decisions.** After seven rounds, treat archetypes as provisional mixes, not labels, and compare any apparent role change with the noise baseline before acting on it.

6.5 Final answer: do the paper's archetypes work for any league?

The method works for any league; the archetypes themselves do not. *Scouting Anyone* makes two claims in one title, and the Liga Femenina separates them.

The method transfers. With nothing but public box scores, the same 15 features and the same settings, Archetypal Analysis finds eight archetypes in a women's league that explain 69.5% of the variation (against 49.6% for K-Means), with a coherence (0.561) between the paper's European (0.572) and NBA (0.538) models, and real players at the corners of six of them. Its core idea, that players are mixtures rather than positions, is even clearer here: 59% of players have no archetype above half their mix. As a scouting tool for a league with little data, it does what the paper promises.

The archetypes do not transfer as a fixed list. Only the role players carry over one to one: Traditional Playmaker ↔ Floor General (+1.66, kept in 100% of variants), 3PT Specialist ↔ Spot-Up Shooter (+1.44, 91%) and High Usage Guard ↔ Scoring Wing (+1.04, 88%). Everything else is reshaped: two men's centre types collapse into one Glass Cleaner, rim protection becomes a specialist's role, the primary creator splits into three, and the Combo Creator appears with no men's counterpart.

The two results are the same result. The archetypes that transfer between leagues are the ones that are stable within a league (Spot-Up Shooter 89%, Floor General 74%), and the ones that do not transfer are the ones that do not even hold within one (Scoring Wing 26%). Role players are defined by a specialty, so the specialty looks the same everywhere; stars are defined by how they create, so they look different

in every league, and on every refit.

The practical answer for any league, then, is not to import the paper's eight labels but to **refit the model on the league itself**: the method will find the role players the paper found, and it will reveal what makes that league's creators and bigs different. For the Liga Femenina Endesa, that is the answer to the title: *any league*, yes, as long as each league is allowed its own archetypes.

7. Limitations and future work

The main limitation is size: one season of 142 stints, against the thousands of player-seasons behind the paper's models.

Limitations

- **Sample size.** Several archetypes are anchored on a single player, so they depend on her presence in the data, and the high-usage archetypes are unstable across refits.
- **Comparison with the paper.** Our numerical profiles are compared with the paper's verbal descriptions; the paper's intra-archetype variance formula is not published; and its examples of mixture weights are three selected players, not full distributions.
- **Choice of K.** No single diagnostic selects K = 8 in this league; the choice rests on several criteria together, including comparability with the paper.
- **Early-season data.** The 2026/27 projection uses seven rounds, and the noise baseline slightly underestimates noise.
- **Data gap.** The 2025/26 box scores of Movistar Estudiantes miss 372 points.

Future work

1. Refit the model without each anchor player and check whether her archetype survives.
2. Fit K = 9 and K = 10 and test whether the extra archetypes are variations of existing ones, as the paper did for its NBA model.
3. Fit Archetypoid Analysis proper, so that every archetype is a real player.
4. Add lineup data and replicate the paper's on/off net-rating analysis for a Liga Femenina team.
5. Pool several seasons to reach a sample closer to the paper's.
6. Ask the paper's authors for their numerical archetype profiles, to replace the verbal encoding with an exact comparison.

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Code and notebooks: 01_dados, 02_model and 03_results (Python, pandas, scikit-learn, archetypes, Plotly).
GitHub: https://github.com/tomascosta7/any-league-archetypes